

Chapter 6: Substances in Daily Use

Textbook Exercise Solutions & Answers



1. Fill in the blanks using proper words:

(a) Rubber made by vulcanization is a material.

hard

(b) Man-made materials are made by natural materials.

processing

(c) thread was developed simultaneously in New York and London.

Nylon

(d) Rayon is also known as

synthetic silk

2. Answer the following questions.

(a) Why did the need for man-made materials arise?

The need for man-made materials arose due to the following reasons:

1. To meet the needs of an increasing population.
2. Human nature to try to make life more comfortable.
3. They can be made available in plenty at a low cost.
4. The reserve of natural substances is decreasing.

c

(b) Which are the natural materials obtained from plants and animals?

Leather, jute, wool, cotton, silk are the natural substances obtained from plants and animals.

(c) What is vulcanization?

1. Vulcanization is the process in which rubber is heated with sulphur for three to four hours.
2. Sulphur is mixed to give hardness to rubber.
3. The proportion of sulphur depends on the purpose for which the rubber is to be used.

(d) Which natural materials are used to obtain fibres?

Cotton, wood pulp and various hydrocarbons obtained from mineral oils are used to obtain fibres.

3. What are we used for?

1. Soil: It supports plant life and hence indirectly supports all living things. It is used for making clay pot, utensils, bricks etc.
2. Wood: It is used in paper industry. It is also used to make furniture.
3. Nylon: It is used to manufacture clothes, fishing nets, ropes, etc.
4. Paper. It is used in our textbooks, note books, currency notes, etc.
5. Rubber: It is used in the manufacture of erasers, tyres, rubber toys, rubber bands, etc.

4. How is paper manufactured? Write in your own words.

Coniferous trees like pine trees are used to make paper.

1. The bark of the logs of these trees is first removed and the wood is broken into small pieces.
2. The mixture of these pieces with some chemicals is kept soaked for a long time to form pulp.
3. On completion of chemical process, fibrous substances from wood pulp are separated and some dyes are added.
4. The pulp is then passed through rollers, dried to form paper and finally wound on reels.

5. Give scientific reasons.

(a) We must use cotton clothes during summer.

1. During summer we sweat more due to high temperature.
2. Cotton clothes absorb sweat.
3. Synthetic clothes are water repellent. They do not absorb sweat and we feel uncomfortable. Hence we must use cotton clothes in summer.

(b) We must observe economy in the use of materials.

1. Due to excessive use of natural substances by human beings to fulfil their needs; they are getting depleted at a faster rate.
2. At the same time, it takes a very long time for these substances to get naturally formed again.
3. Hence, we must observe economy in the use of materials so that they are available for the future generation also.

(c) Saving paper is the need of the hour.

1. Saving paper means saving trees as wood is used as the raw material to manufacture paper.
2. Trees are natural habitat for many Living things.
3. Trees help in increasing rainfall and water availability. Hence, saving paper helps in saving trees which in turn maintains balance in nature.

(d) Man-made materials have more demand.

1. Man-made substances are waterproof, lightweight and easy for transportation.
Substances in Daily Use
2. They are easier to use and can be made available in plenty at a low cost.
Hence, there is more demand for man-made materials.

(e) Humus is a natural material.

1. Humus is obtained from plant and animal wastes.
2. Micro-organisms act on these wastes and convert them into humus.
3. Hence, humus is a natural material.

6. Find out.

(a) How is lac obtained from nature?

1. Lac is a resinous substance secreted from the glands present in the skin of female lac insect.
2. Lac insects live on the Palash trees. In India lac is mainly produced in the states of Rajasthan and Bihar.

(b) How are pearls obtained?

1. Pearls are formed when a foreign particle such as a grain of sand or a small particle of rock accidentally enters the space between the mantle and shell of an oyster's body.
2. Oysters cannot reject the particle, and as a defence mechanism it produces a shining coating called nacre on the particle layer by layer.
3. As the shiny layers get added, a pearl is formed.
4. Cultured pearls are artificially formed by inserting a bead in oyster shell and allowed to coat it with nacre over several years.